



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 21/07/2022*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A + W[n]$, for $n=0, \dots, N-1$, where A is the constant signal of interest (SOI), modelled as a standard Gaussian random variable and $\{W[n]; n = 0, 1, \dots, N-1\}$ are independent random variables modeling additive noise, which is independent of A . Each noise sample is Gaussian distributed, zero mean and with variance σ^2 .

- (1) Derive the covariance matrix $\mathbf{C}_X$ of the observed data vector $\mathbf{X} \triangleq [X[1] \ X[2] \ \dots \ X[N]]^T$ and express it as a function of the signal to noise (SNR) γ defined as: $\gamma \triangleq E\{A^2\} / E\{W^2[n]\}$.
- (2) Apply the Orthogonality Principle and derive the Linear Minimum Mean Square Error (LMMSE) estimator of the random parameter A given the observed data vector $\mathbf{X} \triangleq [X[1] \ X[2] \ \dots \ X[N]]^T$ and express it as a function of the SNR γ .
- (3) Calculate the mean square error (MSE) of the LMMSE estimator and express it as a function of the data size N and of the SNR γ . Plot the MSE as a function of N and as a function of γ and determine whether the estimator is consistent or not.
- (4) Determine numerically the LMMSE and its MSE when $N=20$ and $\sigma^2=2$.
- (5) Derive the Bayesian Cramér-Rao bound (BCRB) on the estimate of the random parameter A and determine whether the LMMSE estimator is efficient or not.
- (6) Can we state that the MAP estimator and the LMMSE estimator coincide in this case?

Woodbury's identity: $(\mathbf{I} + \gamma \mathbf{1}\mathbf{1}^T)^{-1} = \mathbf{I} - \frac{\gamma}{1 + N\gamma} \mathbf{1}\mathbf{1}^T$



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Solution:

$$(1) \quad \mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T, \quad \mathbf{X} | A \in N(\mathbf{A}\mathbf{1}, \sigma^2 \mathbf{I}), \quad A \in N(0, 1),$$

$$\eta_A = E\{A\} = 0, \quad \sigma_A^2 = \text{var}\{A\} = 1,$$

$$E\{X[n]\} = E\{A\} + E\{W[n]\} = 0 \Rightarrow \mathbf{m}_X = E\{\mathbf{X}\} = \mathbf{0}$$

$$\mathbf{C}_X = E\{(\mathbf{X} - \mathbf{m}_X)(\mathbf{X} - \mathbf{m}_X)^T\} = E\{(\mathbf{S} + \mathbf{W})(\mathbf{S} + \mathbf{W})^T\} = E\{(\mathbf{A}\mathbf{1} + \mathbf{W})(\mathbf{A}\mathbf{1} + \mathbf{W})^T\}$$

$$= E\{A^2 \mathbf{1}\mathbf{1}^T\} + E\{\mathbf{W}\mathbf{W}^T\} = \mathbf{C}_S + \mathbf{C}_W = E\{A^2\} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{bmatrix} + \sigma^2 \mathbf{I}$$

$$= \mathbf{1}\mathbf{1}^T + \sigma^2 \mathbf{I} = \sigma^2 \left(\frac{1}{\sigma^2} \mathbf{1}\mathbf{1}^T + \mathbf{I} \right) = \sigma^2 (\gamma \mathbf{1}\mathbf{1}^T + \mathbf{I})$$

$$(2) \quad \hat{A}_{LMMSE} = \sum_{n=0}^{N-1} a_n X[n] + b = \mathbf{a}^T \mathbf{X} + b, \quad \varepsilon = A - \hat{A}_{LMMSE} = A - \mathbf{a}^T \mathbf{X} - b$$

Unbiasedness:

$$E\{\varepsilon\} = E\{A\} - \mathbf{a}^T E\{\mathbf{X}\} - b = 0 \Rightarrow b = E\{A\} - \mathbf{a}^T E\{\mathbf{X}\} = 0$$

Orthogonality Principle:

$$\varepsilon = A - \mathbf{a}^T \mathbf{X}$$

$$E\{\varepsilon \mathbf{X}^T\} = 0 \Rightarrow E\{(A - \mathbf{a}^T \mathbf{X}) \mathbf{X}^T\} = 0$$

$$E\{\varepsilon \mathbf{X}^T\} = E\{A \mathbf{X}^T\} - \mathbf{a}^T E\{\mathbf{X} \mathbf{X}^T\} = \mathbf{c}_{AX} - \mathbf{a}^T \mathbf{C}_X = 0$$

$$\Rightarrow \mathbf{a}^T = \mathbf{c}_{AX} \mathbf{C}_X^{-1} \Rightarrow \mathbf{a} = \mathbf{C}_X^{-1} \mathbf{c}_{AX}^T = \mathbf{C}_X^{-1} \mathbf{c}_{XA}$$

$$\mathbf{c}_{AX} \triangleq E\{A \mathbf{X}^T\} = \mathbf{c}_{XA}^T$$

$$\mathbf{c}_{XA} \triangleq E\{\mathbf{X} A\} = \text{cov}\{X[n], A\} \mathbf{1} = \text{var}\{A\} \mathbf{1} = \mathbf{1}$$



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$$\gamma \triangleq \frac{E\{A^2\}}{E\{W^2[n]\}} = \frac{1}{\sigma^2}$$

$$\mathbf{C}_X = \sigma^2 (\gamma \mathbf{1} \mathbf{1}^T + \mathbf{I}) \Rightarrow \mathbf{C}_X^{-1} = \frac{1}{\sigma^2} (\gamma \mathbf{1} \mathbf{1}^T + \mathbf{I})^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I} - \frac{\gamma}{1+N\gamma} \mathbf{1} \mathbf{1}^T \right)$$

$$\begin{aligned} \mathbf{a} &= \mathbf{C}_X^{-1} \mathbf{c}_{XA} = \frac{1}{\sigma^2} \left(\mathbf{I} - \frac{\gamma}{1+N\gamma} \mathbf{1} \mathbf{1}^T \right) \mathbf{1} = \gamma \left(\mathbf{1} - \frac{\gamma}{1+N\gamma} \mathbf{1} \mathbf{1}^T \mathbf{1} \right) = \gamma \left(\mathbf{1} - \frac{N\gamma}{1+N\gamma} \mathbf{1} \right) = \frac{\gamma}{1+N\gamma} \mathbf{1} \\ &= \frac{N\gamma}{1+N\gamma} \cdot \frac{1}{N} \mathbf{1} = \alpha \frac{1}{N} \mathbf{1}, \quad \text{where: } \alpha \triangleq \frac{N\gamma}{1+N\gamma} \end{aligned}$$

$$\hat{A}_{LMMSE} = \alpha \frac{1}{N} \mathbf{1}^T \mathbf{X} = \alpha \hat{A}_{ML} + (1-\alpha) \eta_A = \alpha \hat{A}_{ML} \quad \text{since } \eta_A = 0$$

where $\hat{A}_{ML} \triangleq \frac{1}{N} \mathbf{1}^T \mathbf{X} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$ is the ML estimator of A in AWGN

(3) MSE

$$\begin{aligned} MSE\{\hat{A}_{LMMSE}\} &= E\{\varepsilon^2\} = E\{\varepsilon A\} = E\left\{ \left(A - \alpha \frac{1}{N} \mathbf{1}^T \mathbf{X} \right) A \right\} = E\{A^2\} - \alpha \frac{1}{N} \mathbf{1}^T E\{\mathbf{X}A\} \\ &= 1 - \alpha \frac{1}{N} \mathbf{1}^T \mathbf{c}_{XA} = 1 - \frac{\alpha}{N} \mathbf{1}^T \mathbf{1} = 1 - \alpha = \frac{1}{1+N\gamma} \end{aligned}$$

(4) For $N = 20$ and $\sigma^2 = 2 \Rightarrow \gamma = \frac{1}{2} \Rightarrow \alpha = \frac{20/2}{1+20/2} = \frac{10}{11} = 0.9091$

$$\hat{A}_{LMMSE} = \frac{10}{11} \cdot \frac{1}{20} \mathbf{1}^T \mathbf{X} = \frac{1}{22} \mathbf{1}^T \mathbf{X} = 0.0455 \cdot \mathbf{1}^T \mathbf{X} = 0.0455 \sum_{n=0}^{19} X[n]$$

$$MSE\{\hat{A}_{LMMSE}\} = \frac{1}{1+N\gamma} = \frac{1}{1+20 \cdot \frac{1}{2}} = \frac{1}{11} = 0.0909$$

(5) BCRB

$$MSE\{\hat{A}\} \geq BCRB(A) = - \frac{1}{E_{A,\mathbf{x}} \left\{ \frac{\partial^2 \ln f_{A,\mathbf{x}}(a, \mathbf{x})}{\partial a^2} \right\}}$$



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where $\ln f_{A,\mathbf{X}}(a, \mathbf{x}) = \ln f_{\mathbf{X}|A}(\mathbf{x}|a) + \ln f_A(a)$, $\mathbf{X}|A \in N(\mathbf{A}\mathbf{1}, \sigma^2\mathbf{I})$, $A \in N(0,1)$

$$\ln f_{\mathbf{X}|A}(\mathbf{x}|a) + \ln f_A(a) = -\frac{(\mathbf{X} - a\mathbf{1})(\mathbf{X} - a\mathbf{1})^T}{2\sigma^2} - \frac{a^2}{2}$$

$$\frac{\partial^2 \ln f_{A,\mathbf{X}}(a, \mathbf{x})}{\partial a^2} = -\frac{N}{\sigma^2} - 1 \Rightarrow BCRB(A) = \frac{1}{\frac{N}{\sigma^2} + 1} = \frac{1}{1 + N\gamma}$$

$$MSE\{\hat{A}_{LMMSE}\} = BCRB(A) = \frac{1}{1 + N\gamma} \Rightarrow \hat{A}_{LMMSE} \text{ is efficient}$$

(6) Since A and $\mathbf{X}$ are jointly Gaussian $MMSE=MAP=LMMSE$